

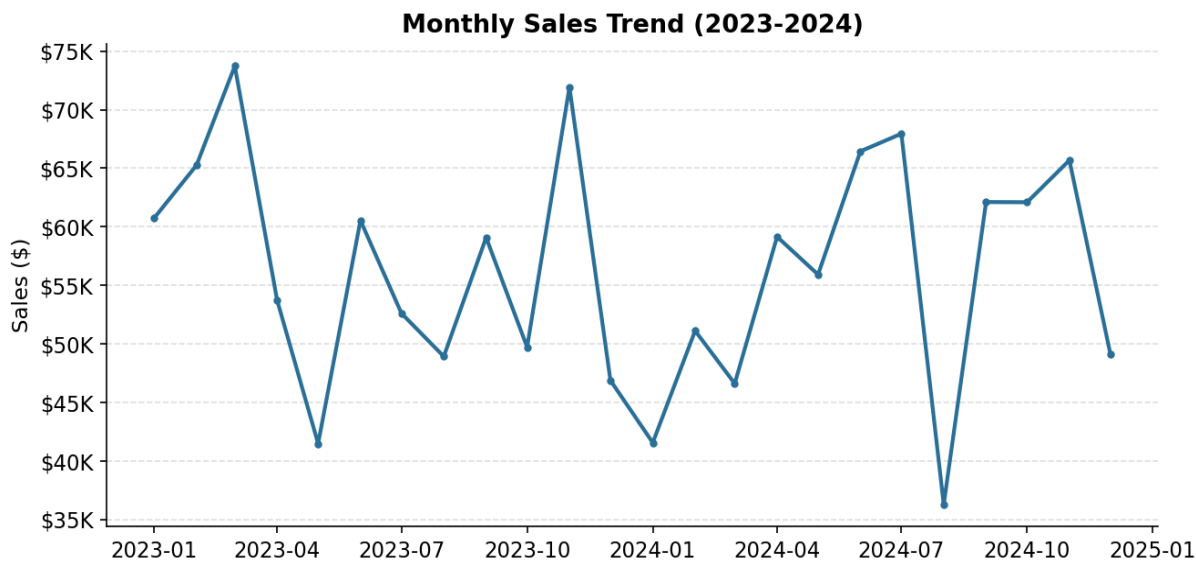
Sales Performance: A Visual Story

Data Analyst Internship - Task 2 | Data Visualization & Storytelling

This report walks through two years (2023-2024) of sales data for a retail business selling Furniture, Office Supplies, and Technology products across the US. The goal isn't just to show charts - it's to answer a business question: **where is the company making money, where is it leaking money, and what should leadership do about it?**

Headline numbers: Total Sales = \$1,348,774 | Total Profit = \$236,015 | Overall margin $\approx$ 17.5% | About 5.5% of all orders were sold at a loss.

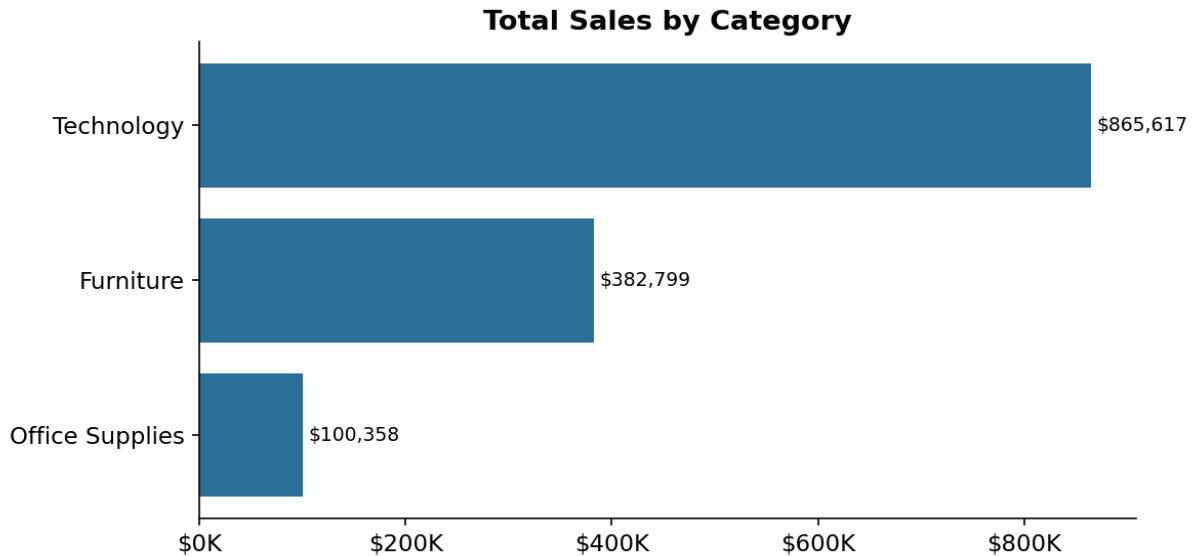
1. Sales Are Growing, But Not Evenly



Monthly sales bounce between roughly \$40K and \$75K, with visible seasonal peaks rather than a flat, predictable line. This kind of month-over-month swing is normal for retail, but it matters for inventory and staffing planning.

Takeaway: Use the high months as a forecasting baseline for promotions and stock-up cycles instead of treating every month the same.

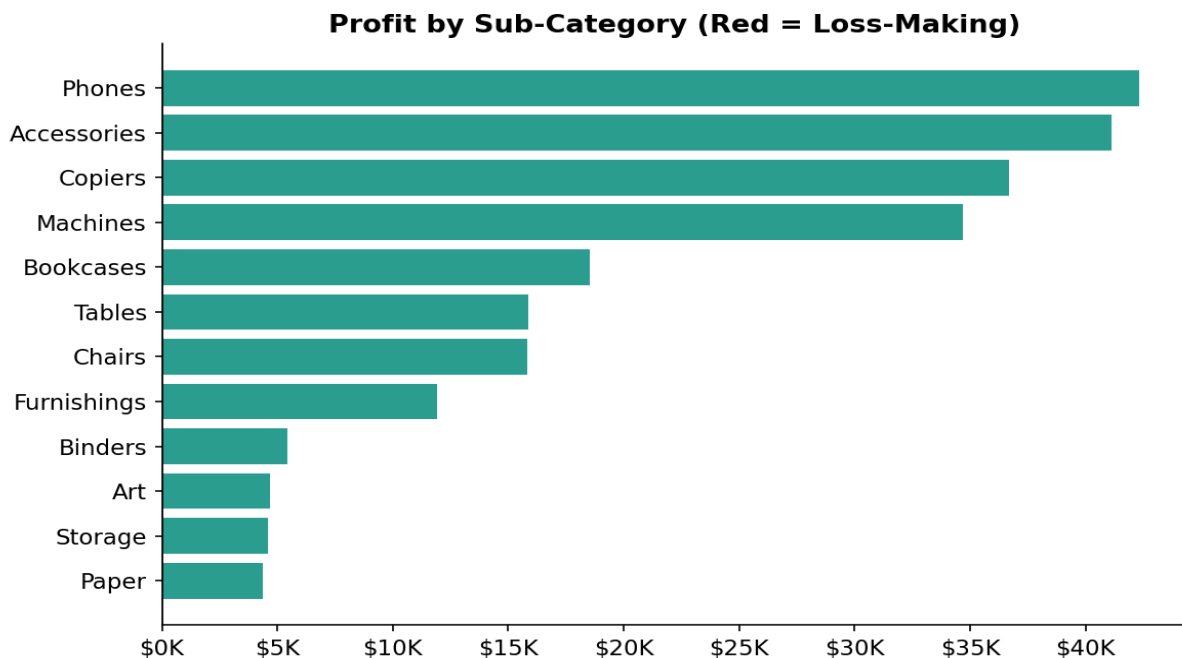
2. Technology Drives Revenue, Office Supplies Barely Registers



Technology alone generated \$865,617 in sales - nearly two-thirds of total revenue - while Office Supplies brought in only \$100,358. A bar chart was used here instead of a pie chart because the goal is to compare exact magnitudes across only three categories, where length is easier to judge accurately than wedge angle.

Takeaway: Technology is the growth engine. Office Supplies may be more of a 'convenience add-on' category than a category worth heavy marketing investment.

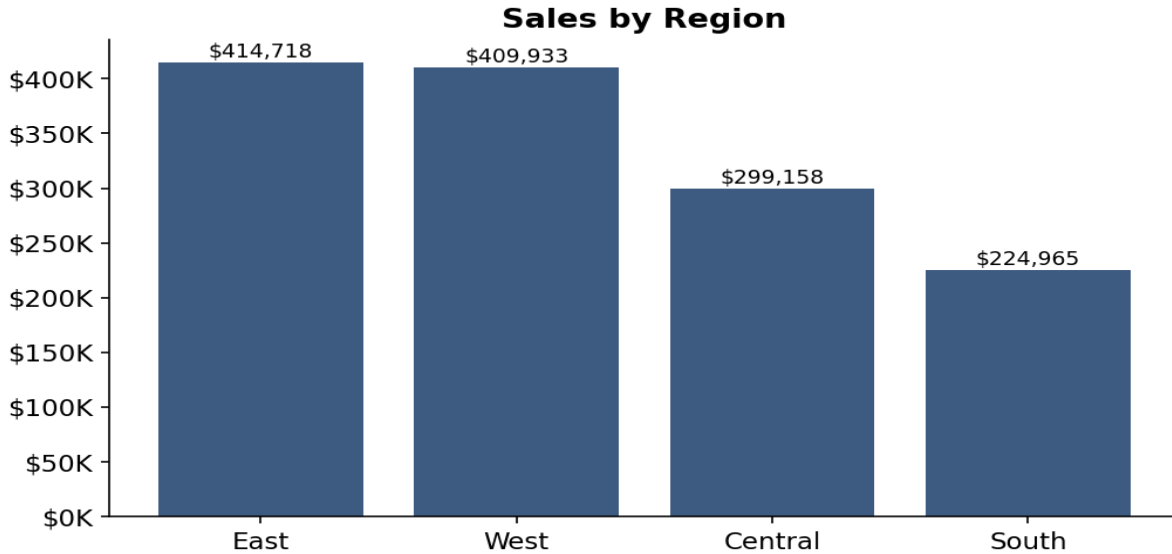
3. Not Every Profitable Category Has Profitable Sub-Categories



Every individual sub-category technically stayed profitable overall, but the gap is stark: Phones and Accessories each generated over \$40K in profit, while Paper, Storage and Art barely cleared \$4-5K. Colour is used here only to flag risk (none currently fall below zero), not for decoration.

Takeaway: Low-margin sub-categories like Paper and Art are good candidates for bundling or reduced promotional spend, freeing budget for high performers like Phones.

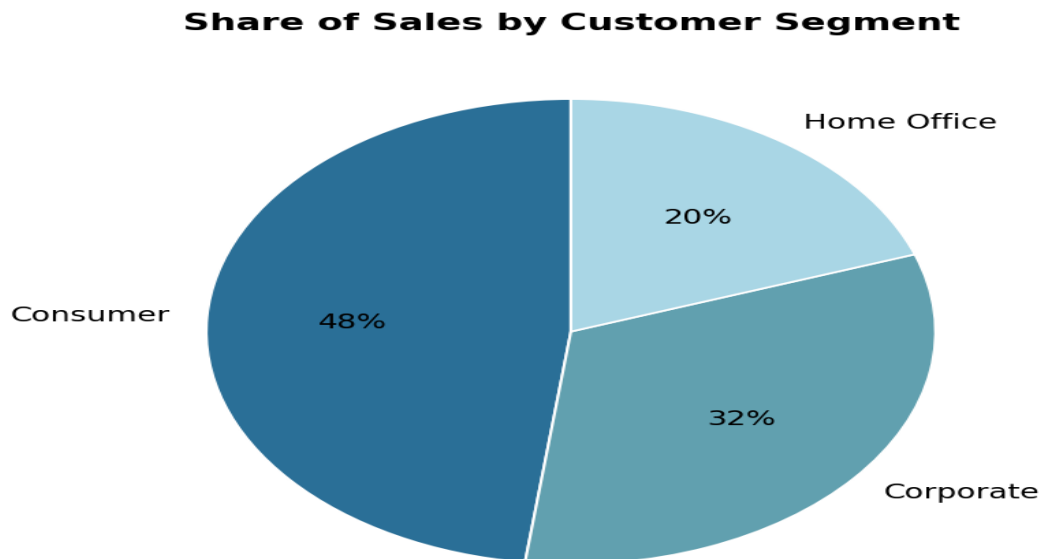
4. East and West Carry the Business



East (\$414,718) and West (\$409,933) regions together account for roughly 61% of all sales, while South lags at \$224,965 - nearly half of East's total.

Takeaway: South region is underperforming relative to its peers and is worth a closer look: is it a smaller market, weaker logistics, or under-investment in marketing?

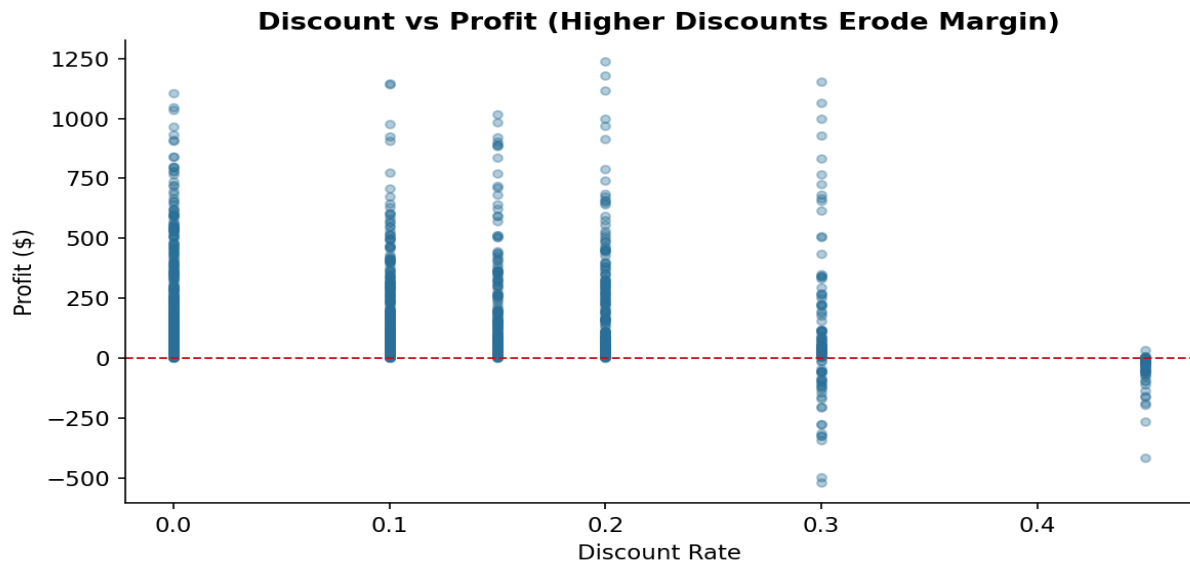
5. Consumers Are the Core Customer Base



A pie chart is appropriate here because the story is about share of a whole (100%) across just three segments. Consumer customers make up about 48% of sales, ahead of Corporate (32%) and Home Office (20%).

Takeaway: Marketing and loyalty programs aimed at individual consumers will likely have the broadest revenue impact, while Corporate accounts may offer room to grow through account-based sales outreach.

6. Heavy Discounting Is Where Profit Goes to Die



Orders placed at a loss carried an average discount rate of about 40%, compared to just 10% for profitable orders. The scatter plot makes this relationship visible at a glance: as discount rate climbs past roughly 30%, profit increasingly dips below zero.

Takeaway: Cap discretionary discounts below 25-30% wherever possible, or pair deep discounts with cost controls (e.g. bundling, reduced shipping tier) so they don't quietly erase the margin on the sale.

Summary Storyboard

Insight	What the data shows	Recommended action
Revenue driver	Technology = 64% of sales	Protect & expand Technology inventory and marketing
Underperformer	Office Supplies = only 7% of sales	Reduce SKU range / bundle with other categories
Weak region	South sales trail East/West by ~45%	Audit pricing, logistics, and local marketing in South
Margin leak	Loss-making orders avg. 40% discount	Set a discount ceiling and monitor margin in real time
Core customer	Consumer segment = 48% of sales	Prioritize loyalty/retention programs for consumers

Note: This dataset is a synthetic sample built to mirror the structure of the classic Superstore dataset (Region, Segment, Category, Discount, Profit, etc.) since a paid Tableau/Power BI license wasn't available. Charts were built in Python (pandas + matplotlib) using the same design principles - right chart for the data, minimal clutter, one colour story per chart, and a takeaway for every chart - that apply equally in Tableau or Power BI.